

Total marks – 120

Attempt Questions 1–8

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^1 \frac{e^x}{1+e^x} dx$. 2

(b) By completing the square, find $\int \frac{dx}{\sqrt{x^2 - 6x + 10}}$. 2

(c) Use integration by parts to find $\int x \log_e x dx$. 3

(d) Use the substitution $u = 1 - x$ to evaluate $\int_{-1}^0 \frac{2+x}{\sqrt{1-x}} dx$. 4

(e) (i) Find real numbers a and b such that 2

$$\frac{4x^2 - 5x - 1}{(x-3)(x^2+1)} \equiv \frac{a}{x-3} + \frac{bx+1}{x^2+1}.$$

(ii) Hence find $\int \frac{4x^2 - 5x - 1}{(x-3)(x^2+1)} dx$. 2

Question 2 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z = 1 + i$ and $w = 1 - 2i$. Find, in the form $x + iy$,

(i) zw 1

(ii) $\frac{1}{\bar{w}}$ 1

(b) Sketch the region in the Argand diagram where the inequalities 3

$$|z - 2 + 2i| \leq 2 \text{ and } \frac{-\pi}{4} < \arg z < \frac{-\pi}{6}$$

hold simultaneously

(c) It is given that $3 - i$ is a root of $P(z) = z^3 + rz + 60$, where r is a real number.

(i) State why $3 + i$ is also a root of $P(z)$. 1

(ii) Factorise $P(z)$ over the real numbers. 2

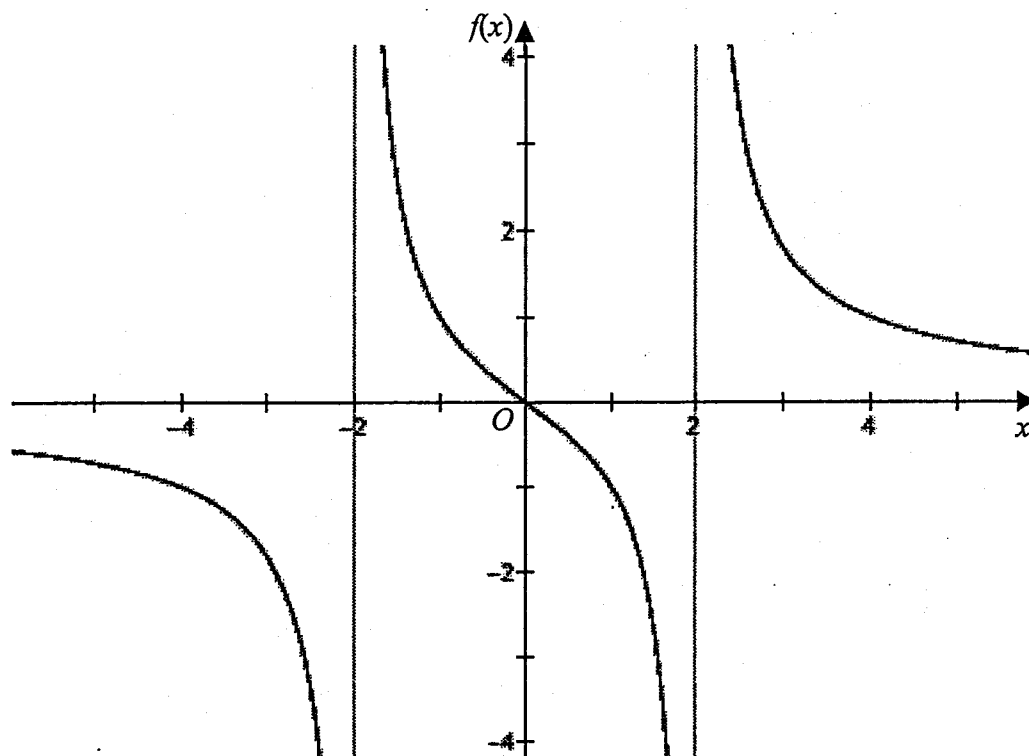
(d) (i) Express $-2 + 2\sqrt{3}i$ in modulus-argument form 2

(ii) Hence evaluate $\sqrt[4]{-2 + 2\sqrt{3}i}$ in the form $x + iy$. 2

(e) By applying de Moivre's theorem and by also expanding $(\cos \theta + i \sin \theta)^4$, 3
obtain expressions for $\cos 4\theta$ in terms of $\cos \theta$.

Question 3 (15 marks) Use a SEPARATE writing booklet.

(a)



The diagram shows the graph of $f(x) = \frac{3x}{x^2 - 4}$.

(i) Write down the equations of all the asymptotes.

1

Draw separate one-third page sketches of the graphs of the following:

(ii) $y = \frac{1}{f(x)}$

2

(iii) $y = f(|x|)$

2

(iv) $y^2 = f(x)$

2

(b) Let α , β and γ be the roots of $x^3 + 3x^2 + 4 = 0$.

(i) Find the polynomial equation whose roots are α^2 , β^2 and γ^2 .

2

(ii) Find $\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3$.

2

Question 3 continues on page 5

Question 3 continued

- (c) (i) If $a > b > 0$, on a sketch of the curve $y = \sqrt{a^2 - x^2}$ shade the region represented by the definite integral $\int_b^a \sqrt{a^2 - x^2} dx$. 1

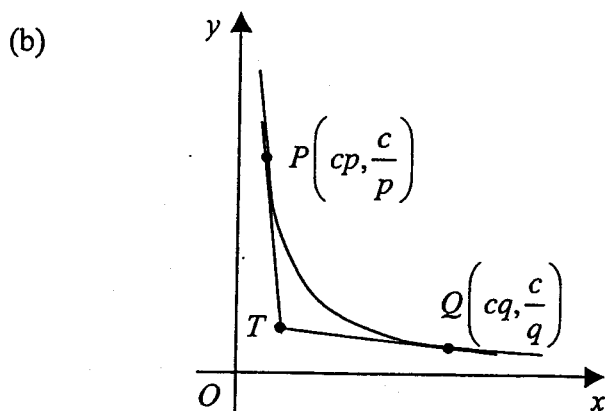
- (ii) By using your diagram, or otherwise, show that 3

$$\int_b^a \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \cos^{-1}\left(\frac{b}{a}\right) - \frac{b}{2} \sqrt{a^2 - b^2}.$$

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the equation of the normal to the curve defined by $x^4 + 3xy - y^2 + 9 = 0$ at the point $(-1, 2)$. 3



The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, $p \neq q$, lie on the same branch of the hyperbola $xy = c^2$.

The tangents at P and Q meet at the point T .

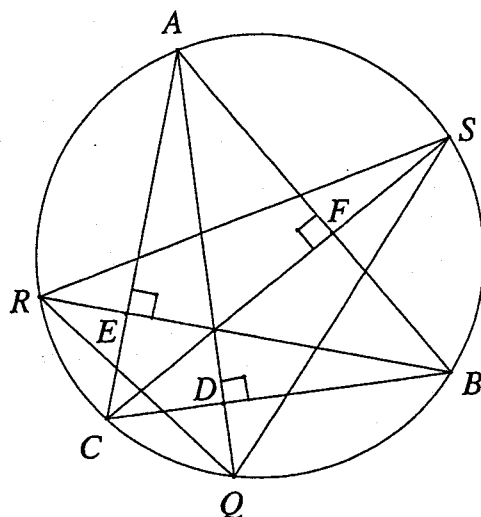
- (i) Show that the equation of the tangent to the hyperbola at Q is $x + q^2y = 2cq$ 2
- (ii) Show that T has coordinates $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$. 2
- (iii) If P and Q move so that $pq = k$, a constant, show that the locus of T is a straight line and give its equation in terms of k . 3
Clearly state any restrictions on the locus.

Question 4 continues on page 7

Question 4 continued

- (c) The vertices of an acute-angled triangle ABC lie on a circle. The perpendiculars from A , B and C meet BC , AC and AB at D , E and F respectively.

The perpendiculars AD , BE and CF are produced to meet the circle at Q , R and S respectively.



- | | | |
|-------|--|---|
| (i) | Using $\triangle AEB$, state the relationship between $\angle EAB$ and $\angle EBA$, giving reasons. | 1 |
| (ii) | Prove that $\angle ABE = \angle ACF$. | 1 |
| (iii) | Prove that AQ bisects $\angle RQS$. | 2 |
| (iv) | Prove that the points E , F , B and C are concyclic. | 1 |

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) The region bounded by $0 \leq x \leq 2$, $0 \leq y \leq 4x^2 - x^4$ is rotated about the y -axis to form a solid.

3

Use the method of cylindrical shells to find the volume of the solid.

- (b) $z_1 = 2\left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right)$ and $z_2 = 2i$ are two complex numbers.

- (i) On an Argand diagram draw the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ to represent z_1 and z_2 respectively. Also draw the vectors representing $z_1 + z_2$ and $z_2 - z_1$.

2

- (ii) Find the exact values of $\arg(z_1 + z_2)$ and $\arg(z_2 - z_1)$.

2

- (c)(i) Show that $f(x) = x\sqrt{4-x^2}$ is an odd function.

1

- (ii) Hence evaluate $\int_{-2}^2 (x-1)\sqrt{4-x^2} dx$

2

- (d)(i) Use the substitution $u = \frac{\pi}{4} - x$ to show that

3

$$\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan x}\right) dx$$

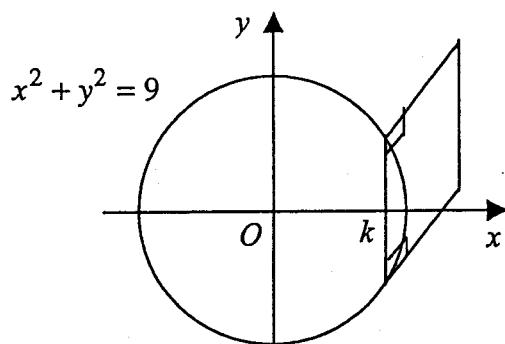
- (ii) Hence find the exact value of $\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx$

2

End of Question 5

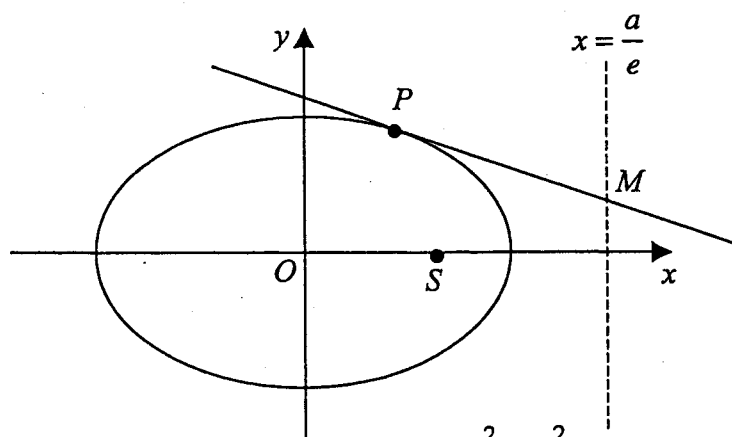
Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) The base of a solid is the region in the xy plane bounded by the circle $x^2 + y^2 = 9$. Each cross-section perpendicular to the x -axis is a square.



- (i) Show that the area of the square cross-section at $x = k$, where $-3 < k < 3$, is $4(9 - k^2)$ 1
- (ii) Hence find the volume of the solid. 2

(b)



The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

S is the focus $(ae, 0)$.

M the point where the tangent at P cuts the directrix, $x = \frac{a}{e}$.

- (i) Show that the equation of the tangent to the ellipse at P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ 2
- (ii) Show that M has coordinates $\left(\frac{a}{e}, \frac{b(e - \cos \theta)}{e \sin \theta} \right)$. 1
- (iii) Prove that $\angle PSM = 90^\circ$. 3

Question 6 continues on page 10

Question 6 continued

- (c) (i) Show that the solutions of $z^6 + z^3 + 1 = 0$ are contained in the solutions of $z^9 - 1 = 0$ 1
- (ii) Sketch the nine solutions of $z^9 - 1 = 0$ on an Argand diagram (about one third of a page in size) 2
- (iii) Mark clearly on your diagram the six roots $z_1, z_2, z_3, z_4, z_5, z_6$ of $z^6 + z^3 + 1 = 0$ 1
- (iv) Show that the sum of the six roots of $z^6 + z^3 + 1 = 0$ is given by $2\left(\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} + \cos\frac{8\pi}{9}\right)$ 2

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) The equation $x^3 - 3x^2 + 5x - 2 = 0$ has roots α, β, γ .

(i) Find $\alpha^2 + \beta^2 + \gamma^2$

1

- (ii) Hence find the number of real roots of the equation
 $x^3 - 3x^2 + 5x - 2 = 0$

2

- (b) A particle of mass m is projected vertically upwards with an initial velocity $u \text{ ms}^{-1}$ in a medium in which the resistance to the motion is proportional to the square of the velocity $v \text{ ms}^{-1}$ of the particle, that is mkv^2 .

Let x be the displacement in metres of the particle above the point of projection, O , so that the equation of motion is

$$\ddot{x} = -(g + kv^2).$$

Where $g \text{ ms}^{-2}$ is the acceleration due to gravity.

- (i) Assume $k = 10$ and that the acceleration due to gravity is 10 ms^{-2} .

Show that $t = \frac{1}{10} (\tan^{-1} u - \tan^{-1} v)$

2

- (ii) Find, in terms of u , the time taken for the particle to reach its greatest height.

1

- (iii) Find an expression for the height, x , in terms of u and v .

2

- (iv) Find, in terms of u , the greatest height attained.

1

- (c) A sequence of numbers u_n is such that $u_1 = 2$, $u_2 = 16$ and
 $u_n = 8u_{n-1} - 15u_{n-2}$ for $n \geq 3$.

- (i) Use the method of Mathematical Induction to show that $u_n = 5^n - 3^n$
for $n \geq 1$.

4

- (ii) Hence show that $u_1 + u_2 + u_3 + \dots + u_n = \frac{5^{n+1} - 2 \times 3^{n+1} + 1}{4}$.

2

Question 8 (15 marks) Use a SEPARATE writing booklet.

(a) Solve $x^x a^{\ln x} = x$ 2

(b) Given that $I_n = \int_0^{\frac{\pi}{4}} \sec x \tan^n x dx$, $n = 1, 2, 3, \dots$

(i) Find I_1 1

(ii) Prove that $I_n = \frac{\sqrt{2}}{n} - \frac{n-1}{n} I_{n-2}$ 3

(iii) Evaluate I_5 1

- (c) Two points, X and Y, are undergoing Simple Harmonic Motion on the number plane.

Y moves along the y axis, oscillating between $y = 1 \text{ cm}$ and $y = 7 \text{ cm}$ with period π seconds.

X moves along the x axis with centre of motion at $x = 3 \text{ cm}$ and period 2π seconds.

Initially Y is stationary at $y = 7$ and X is at its central position with velocity of 2 cms^{-1} .

- (i) Show that Y moves according to the equation $y = 4 + 3 \cos 2t$ and that X moves according to the equation $x = 3 + 2 \sin t$. In both cases you should use the displacement-time equation for SHM, $x = b + a \cos(nt + \alpha)$, as your starting point. 5

- (ii) As X and Y move in SHM find the rate at which the area of triangle OXY is changing when $t = \frac{5\pi}{4}$ 3

End of paper

Question 1

$$\begin{aligned} a. \int_0^1 \frac{e^x}{1+e^x} dx &= \left[\ln(1+e^x) \right]_0^1 \\ &= \ln(1+e) - \ln(1+1) \\ &= \ln\left(\frac{1+e}{2}\right) \end{aligned}$$

$$\begin{aligned} b. \int \frac{dx}{\sqrt{x^2-6x+10}} &= \int \frac{dx}{\sqrt{1+(x-3)^2}} \\ &= \ln(x-3 + \sqrt{(x-3)^2+1}) \\ &= \ln(x-3 + \sqrt{x^2-6x+10}) + C \end{aligned}$$

$$\begin{aligned} c. \int x \ln x \, dx &= \frac{x^2}{2} \ln x - \int \frac{1}{2} \cdot \frac{x^2}{2} dx \\ u &= \ln x \quad v = \frac{x^2}{2} \\ u' &= \frac{1}{x} \quad v' = x \\ &= \frac{x^2}{2} \ln x - \int \frac{x}{2} dx \\ &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C \end{aligned}$$

$$\begin{aligned} d. \int_{-1}^0 \frac{2+x}{\sqrt{1-x}} dx &= - \int_2^1 \frac{3-u}{\sqrt{u}} du \\ u &= 1-x \\ du &= -dx \\ x &= 1-u \\ x=0 &\Rightarrow u=1 \\ x=-1 &\Rightarrow u=2 \\ &= \int_1^2 3u^{-\frac{1}{2}} - u^{\frac{1}{2}} du \\ &= \left[6u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^2 \\ &= 6\sqrt{2} - \frac{2}{3} \cdot 2\sqrt{2} - (6 - \frac{2}{3}) \\ &= \frac{14\sqrt{2}-16}{3} \end{aligned}$$

$$\begin{aligned} e. (i) a(x^2+1) + (bx+1)(x-3) &\equiv 4x^2-5x+1 \\ ax^2+a+bx^2-3bx+x-3 &\equiv 4x^2-5x+1 \\ (a+b)x^2 + (1-3b)x + (a-3) &\equiv 4x^2-5x+1 \\ 1-3b &= -5 \Rightarrow b=2 \\ a-3 &= -1 \Rightarrow a=2 \end{aligned}$$

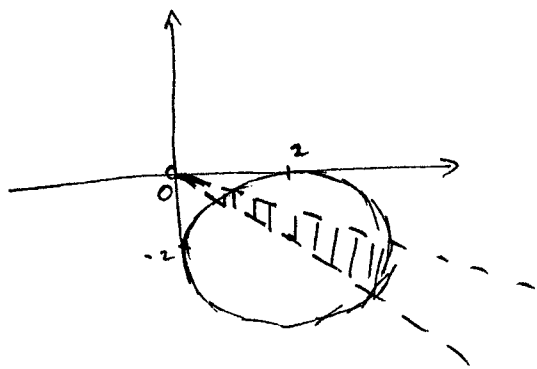
$$\begin{aligned} (ii) \int \frac{4x^2-5x-1}{(x-3)(x^2+1)} dx &= \int \frac{2}{x-3} + \frac{2x+1}{x^2+1} dx \\ &= \int \left(\frac{2}{x-3} + \frac{2x}{x^2+1} + \frac{1}{x^2+1} \right) dx \\ &= 2\ln|x-3| + \ln(x^2+1) + \tan^{-1}x + C \end{aligned}$$

Question 2

$$\begin{aligned} \text{a. (i) } zw &= (1+i)(1-2i) \\ &= 1 - 2i + i - 2i^2 \\ &= 3 - i \end{aligned}$$

$$\begin{aligned} \text{(ii) } \frac{1}{w} &= \frac{1}{(1+2i)} \cdot \frac{(1-2i)}{(1-2i)} \\ &= \frac{1-2i}{1+4} \\ &= \frac{1}{5} - \frac{2}{5}i \end{aligned}$$

b.



(i) Complex roots of polynomials with real coefficients exist as conjugate pairs.

(ii) Roots are $3+i$, $3-i$, α where α real
Using sum of roots $= -\frac{b}{a}$

$$6 + \alpha = 0$$

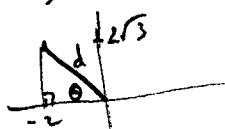
$$\alpha = -6$$

$\therefore z+6$ is a factor

$$\begin{aligned} (z-3-i)(z-3+i) &\text{ is quadratic factor} \\ &= (z-3)^2 - i^2 \\ &= z^2 - 6z + 9 + 1 \\ &= z^2 - 6z + 10 \end{aligned}$$

$$\therefore P(z) = (z+6)(z^2 - 6z + 10)$$

$$\text{d. (i) } -2 + 2\sqrt{3}i$$



$$d^2 = 4 + 12$$

$$= 16$$

$$d = 4$$

$$\tan \theta = \sqrt{3}$$

$$\theta = \frac{\pi}{3}$$

$$\therefore -2 + 2\sqrt{3}i = 4 \operatorname{cis} \frac{2\pi}{3}$$

$$\begin{aligned} \text{(ii) } 4\sqrt{-2+2\sqrt{3}i} &= 4^{\frac{1}{4}} \operatorname{cis} \frac{2\pi}{12} \\ &= \sqrt{2} \operatorname{cis} \frac{\pi}{6} \\ &= \sqrt{2} \left(\frac{\sqrt{3}}{2} + \frac{i}{2} \right) \end{aligned}$$

$$= \frac{\sqrt{6}}{2} + \frac{\sqrt{2}i}{2}$$

$$(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta \quad \text{by De-M.}$$

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= \cancel{\cos^4 \theta} + 4\cos^3 \theta i \sin \theta + 6\cancel{\cos^2 \theta} i^2 \sin^2 \theta \\ &\quad + 4\cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta \\ &= \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta + i(4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta) \end{aligned}$$

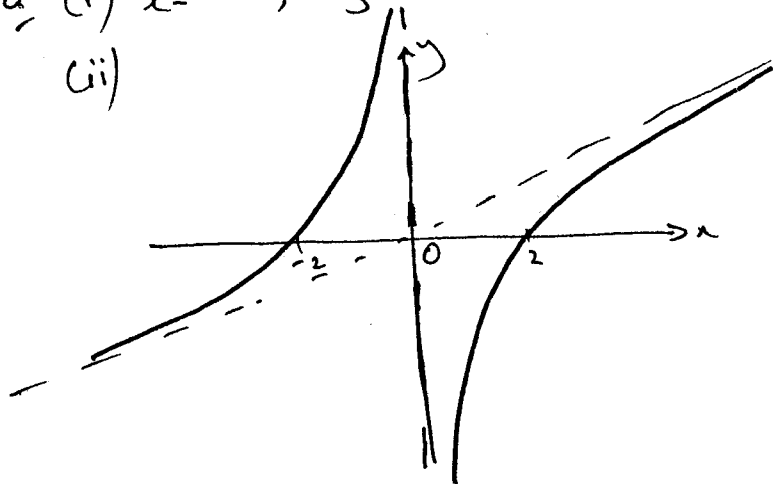
Equating reals

$$\begin{aligned} \cos 4\theta &= \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta \\ &= \cos^4 \theta - 6\cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2 \\ &= \cancel{\cos^4 \theta} - 6\cancel{\cos^2 \theta} + 6\cos^4 \theta + 1 - 2\cos^2 \theta + \cancel{\cos^4 \theta} \\ &= 8\cos^4 \theta - 8\cos^2 \theta + 1 \end{aligned}$$

Question 3

a, (i) $x = \pm 2, y = 0$

(ii)



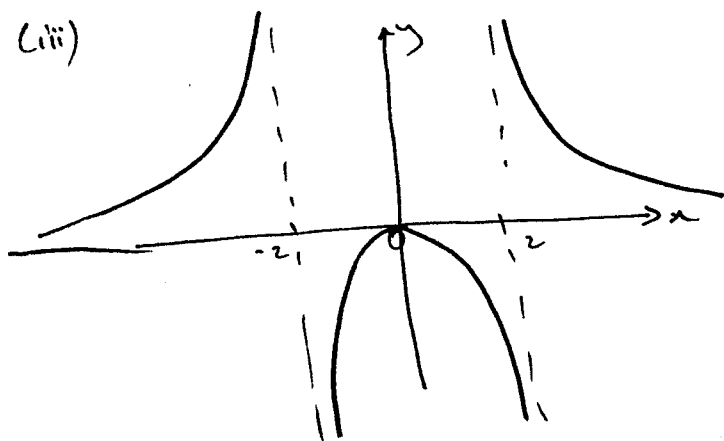
$$\frac{1}{f(x)} = \frac{x^2 - 4}{3x}$$

$$\lim_{x \rightarrow \infty} \frac{x^2 - 4}{3x}$$

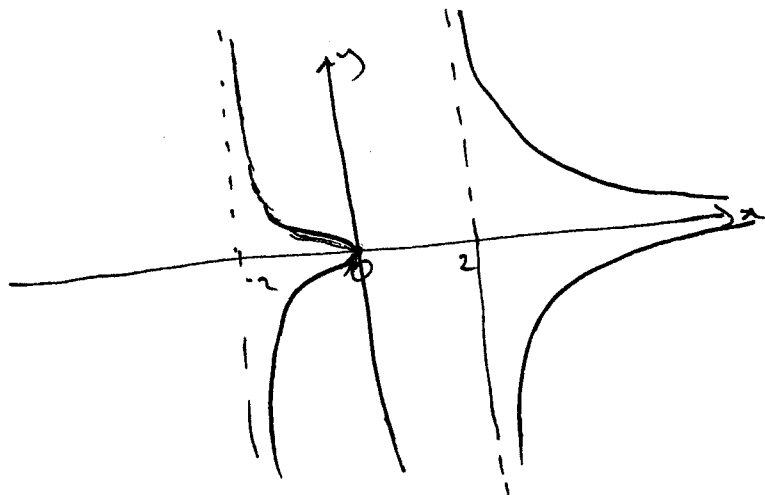
$$= \lim_{x \rightarrow \infty} \frac{x}{3} - \frac{4}{3x}$$

$\therefore$ asymptote $y = \frac{x}{3}$

(iii)



(iv)



Q (i) $(\sqrt{x})^3 + 3(\sqrt{x})^2 + 4 = 0$

$$x\sqrt{x} = -(3x + 4)$$

$$x^3 = (3x + 4)^2$$

$$= 9x^2 + 24x + 16$$

$$x^3 - 9x^2 - 24x - 16 = 0$$

(ii) From (i) $x^2 + y^2 + z^2 = 9$

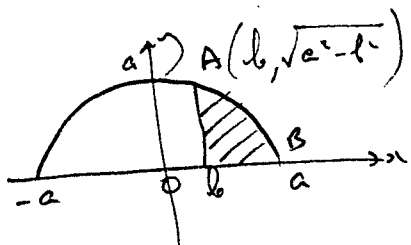
But

$$2xyz = -4$$

$$\therefore x^3 y z + x y^3 z + x y z^3 = x y z (x^2 + y^2 + z^2)$$

$$= -4 \times 9$$

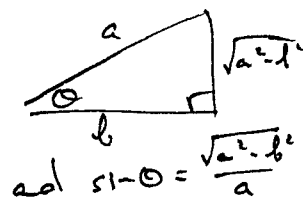
$$= -36$$



(ii) $\int_b^a \sqrt{a^2 - x^2} dx = \text{area sector } OAB - \text{area } \triangle OAB$

$$\cos \theta = \frac{b}{a}$$

Now $\angle AOB = \theta$ such that



$$\therefore \int_b^a \sqrt{a^2 - x^2} dx =$$

$$\frac{1}{2} a^2 \cos^{-1} \left(\frac{b}{a} \right)$$

$$- \frac{1}{2} \cdot b \cdot \sqrt{a^2 - b^2}$$

Question 4

a $x^4 + 3xy - y^2 + 9 = 0$

Diff. w.r. to x

$$4x^3 + 3x \frac{dy}{dx} + 3y - 2y \frac{dy}{dx} = 0$$

$$4x^3 + 3xy = (2y - 3x) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{4x^3 + 3xy}{2y - 3x}$$

For $(-1, 2)$ grad. tangt = $\frac{-4+6}{4+3} = \frac{2}{7}$

$\therefore$ grad normal = $-\frac{7}{2}$

$\therefore$ eqn normal is $y - 2 = -\frac{7}{2}(x + 1)$
 $2y - 4 = -7x - 7$
 $7x + 2y + 3 = 0$

b (i) $xy = c^2$

$$y = c^2 x^{-1}$$

$$y' = -c^2 x^{-2}$$

At Q , grad. tangt = $\frac{-c^2}{c^4 q^2} = -\frac{1}{q^2}$

$\therefore$ eqn tangt is $y - \frac{c}{q} = -\frac{1}{q^2}(x - cq)$
 $q^2 y - cq = -x + cq$
 $x + q^2 y = 2cq$

(ii) $x + p^2 y = 2cp \quad \dots \textcircled{1}$

$x + q^2 y = 2cq \quad \dots \textcircled{2}$

$\textcircled{1} - \textcircled{2} \quad (p^2 - q^2)y = 2c(p - q)$

$$y = \frac{2c}{p+q}$$

Sub into $\textcircled{1} \quad x + \frac{2cp^2}{p+q} = 2cp$

$$(p+q)x + 2cp^2 = 2cp^2 + 2cpq$$

$$x = \frac{2cpq}{p+q}$$

$\therefore T$ is $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$

(ii) At T, with $pq = k$

$$x = \frac{2ck}{p+q}$$

$$y = \frac{2c}{p+q}$$

$$\therefore x = ky$$

$$y = \frac{x}{k} \text{ which is a st. line.}$$

But T is restricted to being "between" the branches of the hyperbola and not including O.

For intersection of $y = \frac{x}{k}$ and $xy = c^2$

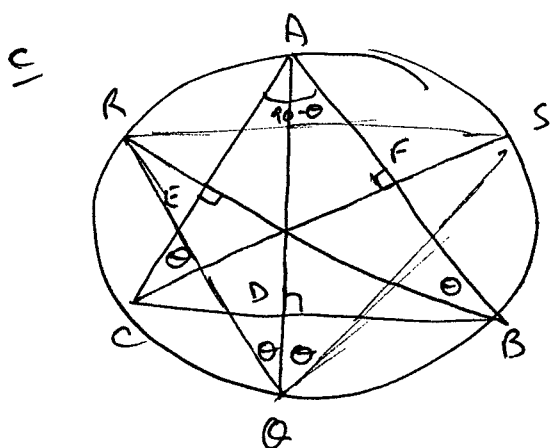
$$\text{get } \frac{x^2}{k} = c^2$$

$$x^2 = c^2 k$$

$$x = \pm c\sqrt{k}$$

So, locus is that part of line $y = \frac{x}{k}$ with domain

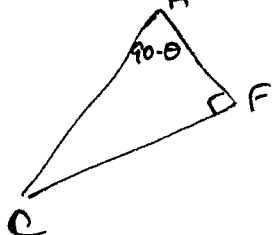
$$-c\sqrt{k} < x < 0, 0 < x < c\sqrt{k}.$$



(i) $\angle EAB = 90 - \angle EBA$

($\angle AEB = 90^\circ \rightarrow BE \perp AC$ (alt))
angle sum of Δ

(ii) $\angle \Delta ACF$



$\therefore \angle ACF = \theta$ (angle sum of Δ)

$\therefore \angle ACF = \angle ABE$

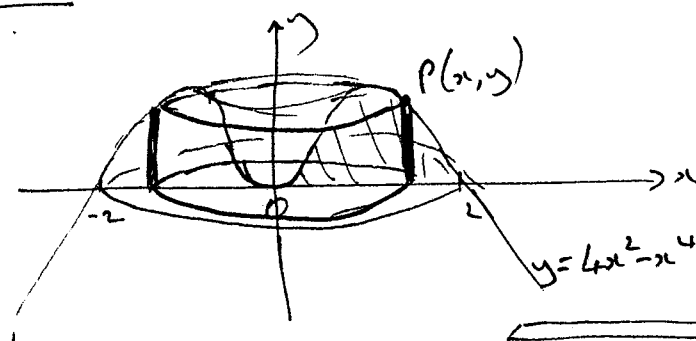
(iii) $\angle AQS = \angle ACF = \theta$ (angles on same arc)
 $\angle RQA = \angle RAB = \theta$ (— — —)

$\therefore \angle RQA = \angle AQS$

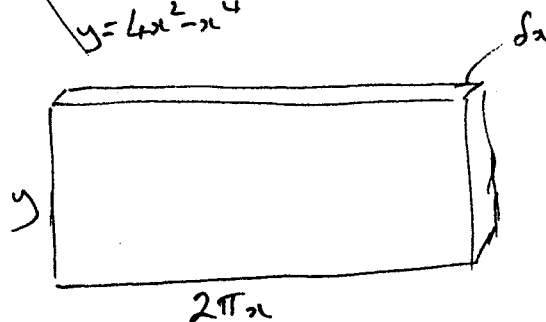
$\therefore AQ$ bisects $\angle ROS$

(iv) $\angle BEC = \angle BFC = 90^\circ$ (alt)
 $\therefore EFBC$ concyclic (BC subtends equal angles at E, F on same side of BC)

Question 5



Take thin-walled hollow cylindrical shell through $P(x, y)$ on curve, thickness δx



$$\text{Vol shell } \delta V = 2\pi xy \delta x$$

$$\text{Total vol} = \lim_{\delta x \rightarrow 0} \sum_{x=0}^2 2\pi xy \delta x$$

$$= 2\pi \int_0^2 xy \, dx$$

$$= 2\pi \int_0^2 (4x^3 - x^5) \, dx$$

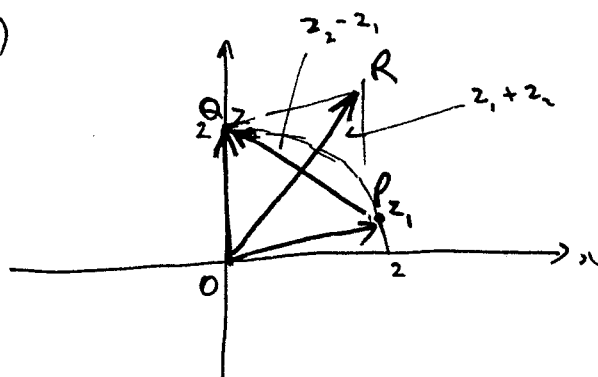
$$= 2\pi \left[x^4 - \frac{x^6}{6} \right]_0^2$$

$$= 2\pi \left[16 - \frac{64}{6} - (0-0) \right]$$

$$= 2\pi \cdot \frac{16}{3}$$

$$= \frac{32\pi}{3} \text{ cu. units}$$

6 (i)



$\vec{OR}$ represents $z_1 + z_2$ - 1
 $\vec{PQ}$ represents $z_2 - z_1$ - 1

$$(ii) \angle xOP = \frac{\pi}{12}$$

$$\angle POQ = \frac{\pi}{2} - \frac{\pi}{12} = \frac{5\pi}{12}$$

Now $OPRQ$ is a rhombus (parallelogram with $OP = OQ$)

$\therefore OR$ bisects $\angle POQ$

$$\therefore \angle POR = \frac{5\pi}{24}$$

$$\therefore \angle xOR = \frac{\pi}{12} + \frac{5\pi}{24} = \frac{7\pi}{24}$$

$$\therefore \arg(z_1 + z_2) = \frac{7\pi}{24} - 1$$

$$(iii) \text{arg. of } z_1 - z_2 = \arg(z_2 - z_1) = \frac{7\pi}{24} + \frac{\pi}{2} = \frac{19\pi}{24} - 1$$

$$f(x) = x\sqrt{4-x^2}$$

$$f(-x) = -x\sqrt{4-(-x)^2}$$

$$= -x\sqrt{4-x^2}$$

$$= -f(x)$$

$f(x) = x\sqrt{4-x^2}$ is an odd function

$$(ii) \int_{-2}^2 (x-1)\sqrt{4-x^2} dx = \int_{-2}^2 x\sqrt{4-x^2} dx - \int_{-2}^2 \sqrt{4-x^2} dx$$

$$= 0 \quad - \text{area of semi-circle of radius 2}$$

$$= -\frac{\pi}{2} \cdot 2^2$$

$$= -2\pi$$

$$d) (i) \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx = -\int_{\frac{\pi}{4}}^0 \ln(1+\tan(\frac{\pi}{4}-u)) du \quad -1$$

$$u = \frac{\pi}{4} - x \quad \therefore x = \frac{\pi}{4} - u$$

$$dx = -du$$

$$\text{Limits } x = \frac{\pi}{4} \Rightarrow u = 0$$

$$x = 0 \Rightarrow u = \frac{\pi}{4}$$

$$= \int_0^{\frac{\pi}{4}} \ln \left[1 + \frac{\tan \frac{\pi}{4} - \tan u}{1 + \tan \frac{\pi}{4} \tan u} \right] du$$

$$= \int_0^{\frac{\pi}{4}} \ln \left[1 + \frac{1 - \tan u}{1 + \tan u} \right] du \quad -1$$

$$= \int_0^{\frac{\pi}{4}} \ln \left[\frac{1 + \tan u + 1 - \tan u}{1 + \tan u} \right] du$$

$$= \int_0^{\frac{\pi}{4}} \ln \left(\frac{2}{1 + \tan u} \right) du$$

$$= \int_0^{\frac{\pi}{4}} \ln \left(\frac{2}{1 + \tan x} \right) dx \quad -1$$

$$(ii) \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx = \int_0^{\frac{\pi}{4}} \ln 2 - \ln(1+\tan x) dx$$

$$= \int_0^{\frac{\pi}{4}} \ln 2 dx - \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx \quad -1$$

$$\therefore 2 \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx = [\ln 2 \cdot x]_0^{\frac{\pi}{4}}$$

$$\int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx = \frac{1}{2} \cdot \frac{\pi}{4} \ln 2$$

$$= \frac{\pi}{8} \ln 2 \quad -1$$

Question 6

2(i) For X-section at $x=k$, base is $2y = 2\sqrt{9-k^2}$

$$\therefore \text{area of X-section} = 4(9-k^2) \quad -1$$

(ii) Vol. X-section slice $\delta V = 4(9-x^2)\delta x$

$$\text{Total vol } V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^3 4(9-x^2)\delta x$$

$$= 8 \int_0^3 (9-x^2) dx \quad -1$$

$$= 8 \left[9x - \frac{x^3}{3} \right]_0^3$$

$$= 8 [27 - 9 - (0-0)] \quad -1$$

$$= 144 \text{ sq. units}$$

b (i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Diff. w.r. to x

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y} = -\frac{b^2 x}{a^2 y}$$

$$\text{At P, grad. of tangent} = -\frac{b^2 a \cos \theta}{a^2 b \sin \theta} = -\frac{b \cos \theta}{a \sin \theta} \quad -1$$

$$\text{Grad. of tangent is } y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a \sin \theta \cdot y - a b \sin^2 \theta = -b \cos \theta \cdot x + a b \cos^2 \theta$$

$$b \cos \theta \cdot x + a \sin \theta \cdot y = a b (\cos^2 \theta + \sin^2 \theta)$$

$$b \cos \theta \cdot x + a \sin \theta \cdot y = a b$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad -1$$

(ii) At M, $x = \frac{a}{e}$ and $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$\frac{y \sin \theta}{b} = 1 - \frac{\cos \theta}{e} = \frac{e - \cos \theta}{e}$$

$$y = \frac{b(e - \cos \theta)}{e \sin \theta}$$

$$\therefore M \text{ is } \left(\frac{a}{e}, \frac{b(e - \cos \theta)}{e \sin \theta} \right) \quad -1$$

$$1) \text{ grad PS} = \frac{b \sin \theta}{a \cos \theta - ae}$$

$$\text{grad. SM} =$$

-1

$$\frac{b(e - \cos \theta)}{\frac{e \sin \theta}{\frac{a}{e} - ae}}$$

-1

$$\text{grad PS} \cdot \text{grad SM} =$$

$$= \frac{b \sin \theta}{a(\cos \theta - e)} \cdot \frac{b(e - \cos \theta)}{\frac{e \sin \theta}{\frac{a}{e} - ae}} \cdot \frac{e}{a(1 - e^2)}$$

$$= \frac{b^2}{-a^2(1 - e^2)}$$

$$= \frac{b^2}{-b^2}$$

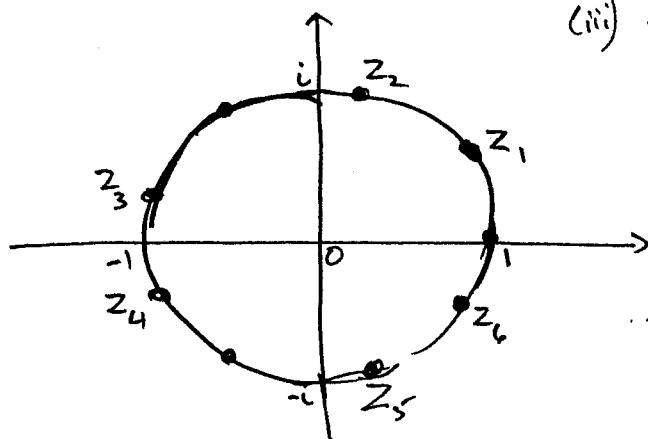
$$= -1$$

-1

$$\therefore \text{PS} \perp \text{SM}$$

$$\text{c (i) } z^9 - 1 = (z^3)^3 - 1 = (z^3 - 1)(z^6 + z^3 + 1) \quad -1$$

(ii)



$$\text{(iii) roots of } z^3 - 1 = 0 \text{ are } 1, \text{cis } \frac{2\pi}{3}, \text{cis } -\frac{2\pi}{3}$$

$$\text{other roots are for } z^6 + z^3 + 1 = 0$$

$$2 \begin{cases} 1 & \text{for } r=1 \\ 1 & \text{for } \frac{2\pi}{9} \text{ diff.} \end{cases}$$

$$\begin{aligned} \text{(iv) } z_1 + z_2 + z_3 + z_4 + z_5 + z_6 &= \text{cis } \frac{2\pi}{9} + \text{cis } \frac{4\pi}{9} + \text{cis } \frac{8\pi}{9} \\ &\quad + \text{cis } (-\frac{2\pi}{9}) + \text{cis } (-\frac{4\pi}{9}) + \text{cis } (-\frac{8\pi}{9}) \\ &= 2 \left(\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} \right) \end{aligned} \quad \begin{matrix} 1 \\ 1 \end{matrix}$$

Question 7

a. $x^3 - 3x^2 + 5x - 2 = 0$

(i) $\alpha + \beta + \gamma = 3$

$\alpha\beta + \alpha\gamma + \beta\gamma = 5$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 9 - 10 = -1$$

(ii) Since $\sum \alpha^2$ is negative, at least one of $\alpha^2, \beta^2, \gamma^2$ is negative
 i.e. at least one of α, β, γ is complex. - 1

But coeff of eqn are real

$\therefore$ complex roots exist as conjugate pairs. - 1

Hence 2 complex roots
 i.e. 1 real root.

b. (i) $\ddot{v} = -(10 + 10v^2)$

$\frac{dv}{dt} = -10(1+v^2)$

$\frac{dt}{dv} = -\frac{1}{10(1+v^2)}$

$t = -\frac{1}{10} \int \frac{dv}{1+v^2}$ - 1

$t = -\frac{1}{10} \tan^{-1} v + c$

When $t=0, v=u \quad \therefore c = \frac{1}{10} \tan^{-1} u$

$\therefore t = \frac{1}{10} (\tan^{-1} u - \tan^{-1} v)$ - 1

(ii) Greatest height when $v=0$

$\therefore t = \frac{1}{10} \tan^{-1} u$ - 1

(iii)

$v \frac{dv}{dt} = -10(1+v^2)$

$\frac{dv}{dt} = -\frac{10(1+v^2)}{v}$

$\frac{dx}{dv} = -\frac{v}{10(1+v^2)}$

$x = -\frac{1}{10} \int \frac{v}{1+v^2} dv$ - 1

$x = -\frac{1}{10} \cdot \frac{1}{2} \ln(1+v^2) + c$

When $x=0, v=u \quad \therefore c = \frac{1}{20} \ln(1+u^2)$

$x = \frac{1}{20} \ln\left(\frac{1+u^2}{1+v^2}\right)$ - 1

(iv) Greatest height when $v=0 \quad \therefore x = \frac{1}{20} \ln(1+u^2)$ - 1

(i) Prove true for $n=1$ & $n=2$

$$u_1 = 5^1 - 3^1$$

$$= 2$$

True

$$u_2 = 5^2 - 3^2$$

$$= 16$$

True

$\therefore$ true for $n=1$ & $n=2$

Assume true for $n=1, 2, \dots, k$

ie assume $u_k = 5^k - 3^k$

$$u_2 = 5^2 - 3^2$$

$$\hat{u}_k = 5^k - 3^k$$

Prove true for $n=k+1$ if true for $n=1, 2, \dots, k$

ie prove $u_{k+1} = 5^{k+1} - 3^{k+1}$

Now $u_{k+1} = 8u_k - 15u_{k-1}$

$$= 8(5^k - 3^k) - 15(5^{k-1} - 3^{k-1}) \quad -1$$

$$= 8 \times 5^k - 3 \times 5 \times 5^{k-1} + 5 \times 3 \times 3^{k-1} - 8 \times 3^k$$

$$= 8 \times 5^k - 3 \times 5^k + 5 \times 3^k - 8 \times 3^k$$

$$= 5 \times 5^k - 3 \times 3^k$$

$$= 5^{k+1} - 3^{k+1} \quad -1$$

Conclusion

True for $n=k+1$ if true for $n=1, 2, \dots, k$

But true for $n=1, 2$

$\therefore$ true for all integer $n \geq 1$

(ii) $u_1 + u_2 + \dots + u_n = 5^1 - 3^1 + 5^2 - 3^2 + \dots + 5^n - 3^n$

$$= (5^1 + 5^2 + \dots + 5^n) - (3^1 + 3^2 + \dots + 3^n) \quad -1$$

= GP $a=5$
 $r=5$

GP $a=3$
 $r=3$

$$= \frac{5(5^n - 1)}{4} - \frac{3(3^n - 1)}{2}$$

$$= \frac{5^{n+1} - 5}{4} - \frac{3^{n+1} - 3}{2}$$

$$= \frac{5^{n+1} - 5 - 2 \times 3^{n+1} + 6}{4}$$

$$= \frac{5^{n+1} - 2 \times 3^{n+1} + 1}{4} \quad -1$$

Question 8

a. $x^a a^{\ln x} = x$

$\therefore \ln x^a a^{\ln x} = \ln x$

$\ln x^a + \ln a^{\ln x} = \ln x$

$x \ln x + \ln x \cdot \ln a = \ln x \quad -1$

$\ln x(x + \ln a - 1) = 0$

$\ln x = 0$ or $x + \ln a - 1 = 0$

$x = 1$ or $x = 1 - \ln a \quad -1$

b. $I_n = \int_0^{\frac{\pi}{4}} \sec x \tan^n x \, dx$

(i) $I_1 = \int_0^{\frac{\pi}{4}} \sec x \tan x \, dx$

$= [\sec x]_0^{\frac{\pi}{4}}$

$= \sec \frac{\pi}{4} - \sec 0$

$= \sqrt{2} - 1 \quad -1$

(ii) $I_n = \int_0^{\frac{\pi}{4}} \sec x \tan^n x \, dx = [\tan^{n-1} x \sec x]_0^{\frac{\pi}{4}}$

$u = \tan^{n-1} x \quad v = \sec x$

$u' = (n-1) \tan^{n-2} x \sec^2 x \quad v' = \sec x \tan x$

$= \int_0^{\frac{\pi}{4}} (n-1) \tan^{n-2} x \sec^2 x \sec x \, dx$

$= \left[\left(\tan \frac{\pi}{4} \right)^{n-1} \cdot \sec \frac{\pi}{4} - 0 \right]$

$-(n-1) \int_0^{\frac{\pi}{4}} \sec x \tan^{n-2} x (1 + \tan^2 x) \, dx$

$= \sqrt{2} - (n-1) \left[\int_0^{\frac{\pi}{4}} \sec x \tan^{n-2} x \, dx - \int_0^{\frac{\pi}{4}} \sec x \tan^n x \, dx \right]$

$I_n = \sqrt{2} - (n-1) [I_{n-2} - I_n] \quad -1$

$= \sqrt{2} - (n-1) I_{n-2} - (n-1) I_n$

$\therefore n I_n = \sqrt{2} - (n-1) I_{n-2}$

$I_n = \frac{\sqrt{2}}{n} - \left(\frac{n-1}{n} \right) I_{n-2} \quad -1$

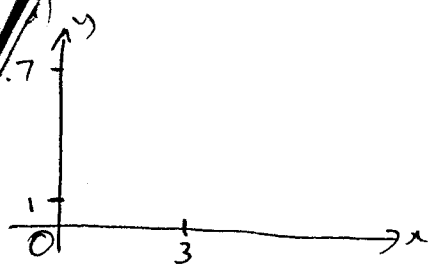
(iii) $I_5 = \frac{\sqrt{2}}{5} - \frac{4}{5} I_3$

$= \frac{\sqrt{2}}{5} - \frac{4}{5} \left[\frac{\sqrt{2}}{3} - \frac{2}{3} I_1 \right]$

$= \frac{\sqrt{2}}{5} - \frac{4\sqrt{2}}{15} + \frac{8}{15} (\sqrt{2} - 1)$

$= \frac{3\sqrt{2}}{15} - \frac{4\sqrt{2}}{15} + \frac{8\sqrt{2}}{15} - \frac{8}{15}$

$= \frac{7\sqrt{2}}{15} - \frac{8}{15} \quad -1$



$\underline{Y}$: centre of motion $y=4$ $\therefore b=4$
 amplitude $a=3$
 period $\frac{2\pi}{\omega} = \pi$ $\therefore \omega = 2$
 Initially at $y=7$ $\therefore \phi = 0$
 $\therefore y = 4 + 3 \cos 2t$

$\underline{X}$: centre of motion $x=3$ $\therefore b=3$
 period $\frac{2\pi}{\omega} = 2\pi$ $\therefore \omega = 1$
 At centre $v^2 = \omega^2 a^2$ $\therefore a=2$
 Initially at centre, moving $\therefore \phi = -\frac{\pi}{2}$



$\therefore x = 3 + 2 \cos(t - \frac{\pi}{2})$
 $= 3 + 2 \cos(\frac{\pi}{2} - t)$
 $= 3 + 2 \sin t$

(ii) Area $\Delta OXY = \frac{1}{2} (4 + 3 \cos 2t) (3 + 2 \sin t)$
 $\frac{dA}{dt} = \frac{1}{2} [(4 + 3 \cos 2t) \times 2 \cos t + (3 + 2 \sin t) \times -6 \sin 2t]$

When $t = \frac{5\pi}{4}$, $\frac{dA}{dt} = \frac{1}{2} [(4 + 3 \cos \frac{5\pi}{2}) \times 2 \cos \frac{5\pi}{4} + (3 + 2 \sin \frac{5\pi}{4}) \times -6 \sin \frac{5\pi}{2}]$
 $= \frac{1}{2} [(4 + 0) \times -\frac{2}{\sqrt{2}} + (3 - \frac{2}{\sqrt{2}}) \times -6]$
 $= \frac{1}{2} [-\frac{8}{\sqrt{2}} - 18 + \frac{12}{\sqrt{2}}]$
 $= \frac{1}{2} [\frac{4}{\sqrt{2}} - 18]$
 $= \frac{2}{\sqrt{2}} - 9$
 $= \sqrt{2} - 9$

$\therefore$ area decreasing at $(9 - \sqrt{2}) \text{ cm}^2 \text{ s}^{-1}$
 or area changing at $(\sqrt{2} - 9) \text{ cm}^2 \text{ s}^{-1}$